

Chapter 3

Design, Fabrication, and Performance of CCD Imagers

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I. INTRODUCTION

A. History

It was recognized almost immediately in 1970, after the charge-coupled device (CCD) principle was first demonstrated [1], that this new development could lead to a significant technology for the manufacture of large

image sensors [2]. At that time there were already considerable interest and activity in image sensor chips that could be used, either singly or in more complex optical systems, for a number of types of applications, including aerial electronic cameras, page scanners, and miniature and/or rugged television-type cameras; these applications are still important today.

A more detailed early history of CCDs is given by Séquin and Tompsett in their 1975 book on charge-transfer devices [3]. In this chapter we shall be concerned more specifically with the development of CCD image sensors in the context of very large scale integration (VLSI) and with the maturing of the technology as evidenced by commercial availability. An important innovation, which was developed shortly after 1970, was the buried-channel concept for the design of CCDs [4,4a]. With buried-channel CCDs (BCCDs), it was soon demonstrated [5] that charge-transfer efficiencies (CTEs) of at least 0.9999 were a possibility at high transfer frequencies, i.e., over 1 MHz. The improvement (decrease in transfer inefficiency) over the surface-channel CCD technology is generally accepted to be at least an order of magnitude. This high CTE has made it practical to consider very large imager designs with 1000 or more stages of serial charge transfer.

B. Scope of This Chapter

To date, numerous comprehensive papers and reviews have been written about CCD imagers [3,6–11], commercially available devices have existed for over seven years [12], and a number of companies in several countries currently make CCD imagers commercially available. It is hoped that the reader will find this chapter to be a significant update and a useful reference for the future.

The objectives of this chapter are to provide the reader with (1) a detailed view of one interesting product area of VLSI device development, (2) a review of the salient developments in several subcategories of CCD imager technology and product development, (3) a summary of the fundamental limitations on device performance, and (4) some descriptions and performance information for commercially available devices. Because of the fact that these commercially available devices are primarily line-scan image sensors and two-dimensional arrays with intrinsic silicon spectral response, it is intentional that there will be a heavy emphasis on these types of devices. Beyond the scope of this chapter are hybrid focal planes and a thorough treatment of the wafer-processing technology.

C. CCD Imagers as Prime Examples of VLSI Structures

Silicon image sensors have, since early in the development of CCD technology, been leading examples of the larger LSI and VLSI structures. For example, in 1973, the Bell Telephone Laboratories [13] showed imagery obtained from a line-scan imager with an active length of 22.5 mm (1500 elements on 15- μm centers). In 1974, RCA [14] showed imagery obtained with a frame-transfer-type area image sensor with $512 \times 320 = 163,840$ interlaced sense elements and half that many, 81,920, storage elements. The chip size of the latter device is $12.7 \times 19.0 = 241 \text{ mm}^2$. In 1978, Texas Instruments [15] showed imagery obtained with an 800×800 -element “snap-shot”-type imager. This device, with its 15- μm square elements, has an active imaging area of 144 mm^2 and a total chip area of $17.8 \times 17.8 = 317 \text{ mm}^2$. A photograph of the chip is shown in Fig. 1. (The wide margin is needed for support of the back-side-thinned imaging portion of the device.)

Similar CCD image sensors have also been leading examples of VLSI in

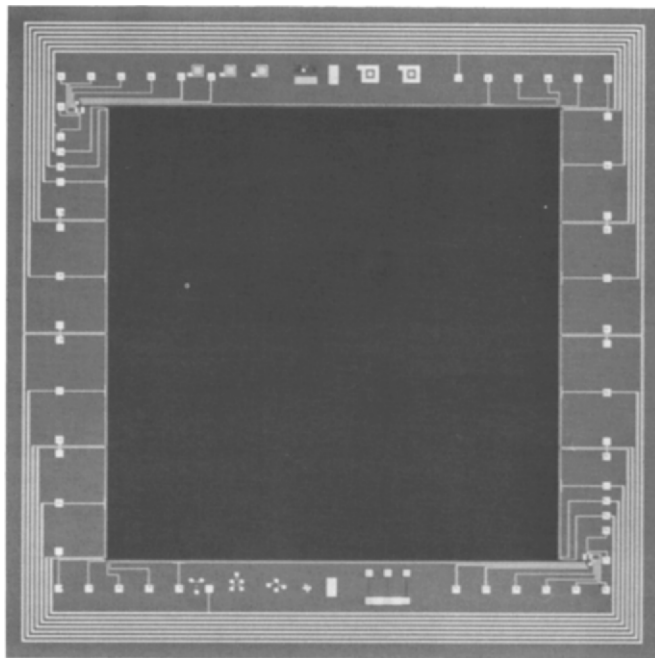


Fig. 1. Texas Instruments's 800×800 -element “snap-shot” imager (courtesy of Texas Instruments).

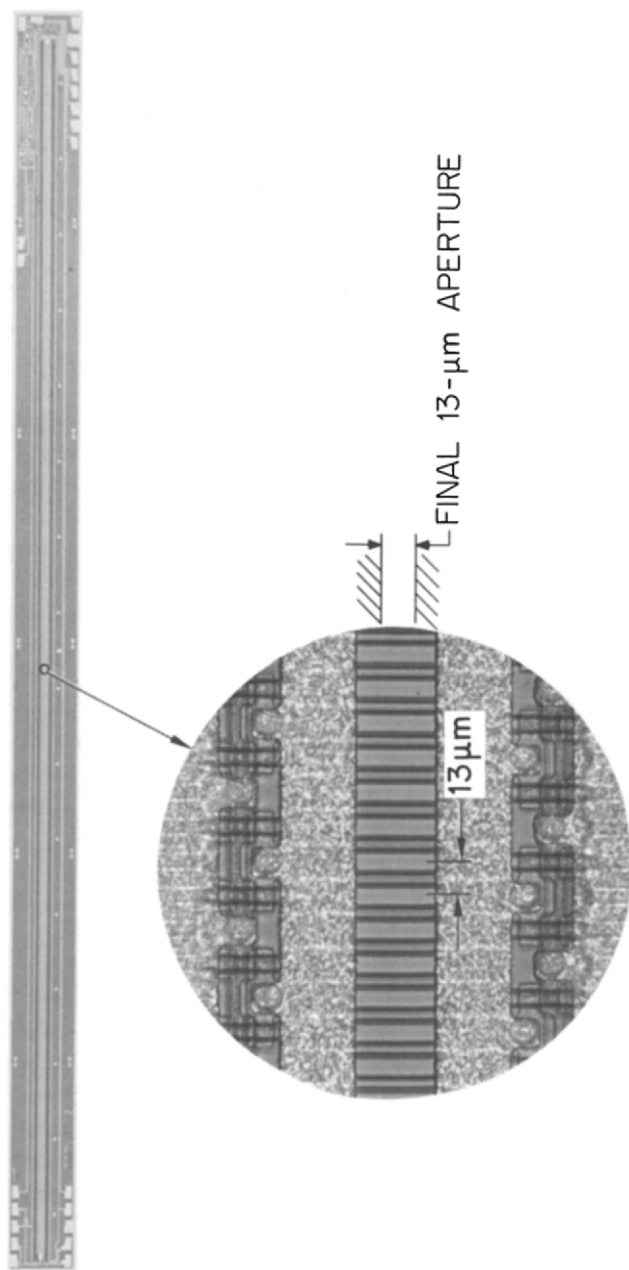


Fig. 2. The Fairchild CCD142 line-scan imager having 2048 photoelements and a chip length of 27.6 mm. (The second layer of metallization has been omitted for this photograph in order to show more detail.)

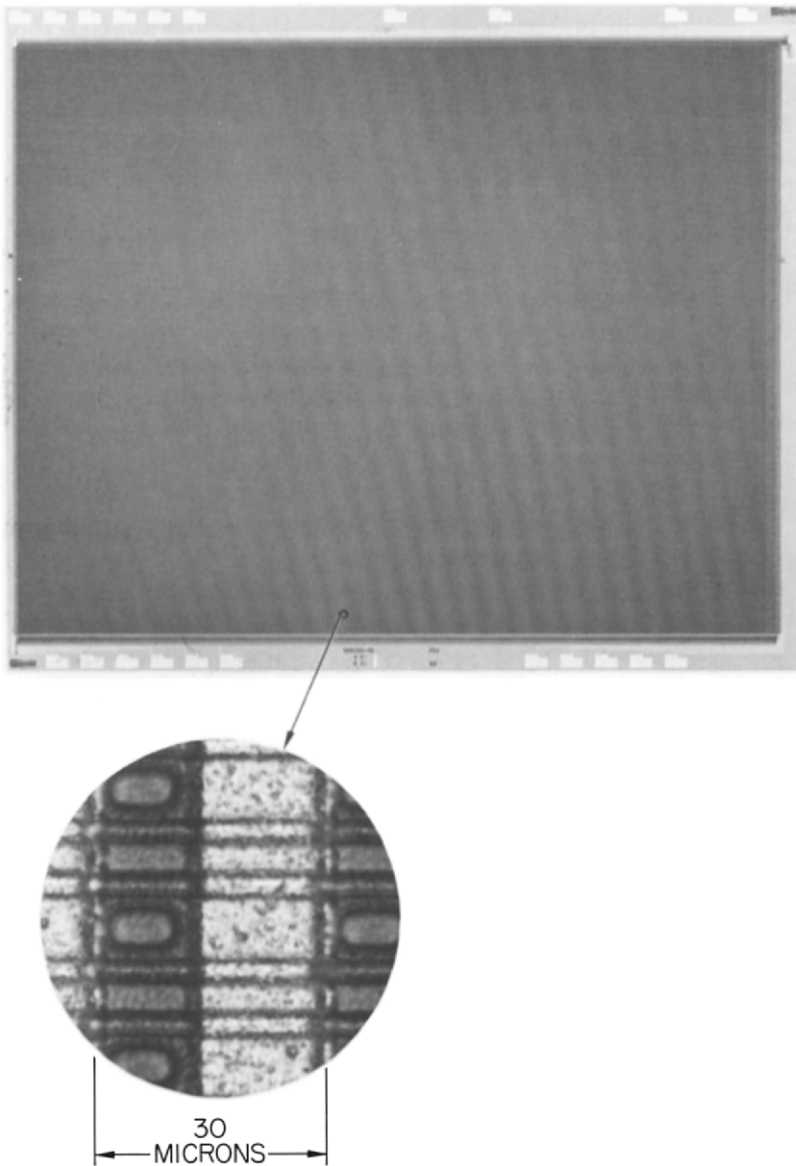


Fig. 3. The Fairchild CCD221 interline transfer imager having 488×380 photoelements and a chip area of 122 mm^2 .

their commercial availability. A 500-element line-scan imager¹ [12,16] was announced in 1973 as the first commercially available CCD. It had an active length of $30.48 \mu\text{m} \times 500 = 15.2 \text{ mm}$. In 1975, a 1728-element line-scan imager² was announced [5,17], with active length $13 \mu\text{m} \times 1728 = 22.5 \text{ mm}$. In that same year, RCA brought out its "Big SID" (Silicon Imaging Device) [14,18] as an available component—the 512×320 -element frame-transfer device previously described.

Photographs of two currently commercially available devices are shown in Figs. 2 and 3. The first is a 2000-element line-scan imager designated as the Fairchild CCD142. Its active length is 26.0 mm and its chip length is 27.6 mm. This device will be described in more detail in Section II. Figure 3 shows a 488×380 -element interline transfer (ILT) imager designated as the Fairchild CCD221. The active sensing area of this device is $11.4 \times 8.8 = 100 \text{ mm}^2$ and the chip area is $10.1 \times 12.1 = 122 \text{ mm}^2$. By comparison, the 64,000-bit dynamic RAMs of 1980 have chip areas that range approximately from 20 to 30 mm^2 ; for example, the Fairchild F64K's chip area is 23.5 mm^2 .

II. SILICON CCD IMAGERS WITH INTRINSIC PHOTORESPONSE

A. Fundamentals of Device Performance

1. Internal Responsivity

Intrinsic silicon photodevices can be responsive from the absorption edge near 1100 nm in the infrared to far into the ultraviolet. The internal responsivity is limited near the absorption edge by the absorption coefficient and the minority carrier diffusion length below the depletion layer. Figure 4 shows possible internal responsivity spectra calculated using the model of complete charge collection in the depletion layer (thickness L_D) and exponentially decaying charge collection efficiency below the depletion layer, the latter depending on the minority carrier diffusion length L_n . (A scale of inverse absorption coefficient is shown at the top of the figure.) The figure also indicates the spectral response contribution from the depletion layer alone. This figure will be helpful in the next subsection in understanding how optical cross talk, due to the diffusion component of responsivity, increases with wavelength.

¹ The Fairchild CCD101 [12].

² The Fairchild CCD121 [17].

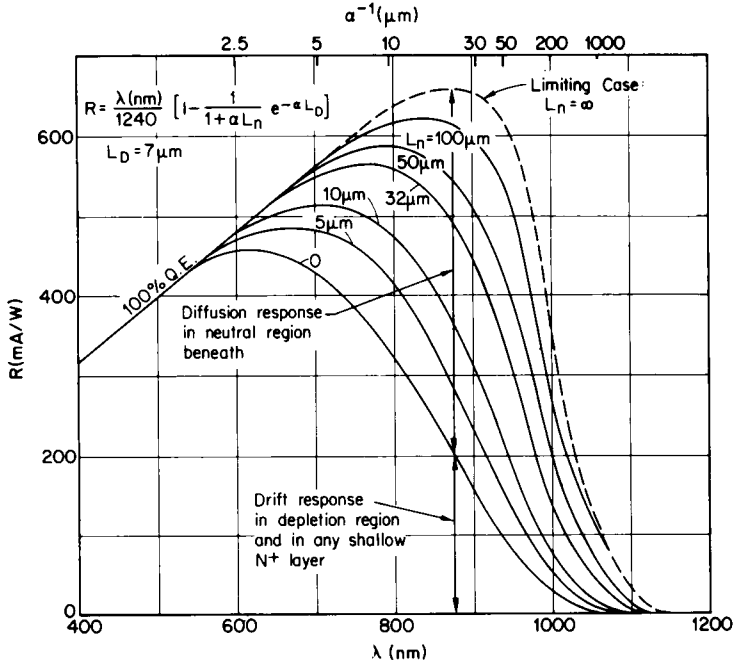


Fig. 4. Theoretical internal spectral response for silicon photosensors in the absence of reflection losses and gate absorption losses. For Fairchild CCD image sensors, the electron diffusion length L_n typically is in the range 10–100 μm . Silicon thickness = 250 μm ; back surface ohmic contact ($S_0 = \infty$); neglects any back surface reflection.

On the short-wavelength end of the spectrum, although silicon detectors have exhibited useful performance at wavelengths as short as 200 nm, it is generally not recommended to irradiate MOS devices or CCDs from the front side at less than approximately 300 nm due to possible instabilities brought about by photoemission into the oxide layer just above the irradiated surface of the silicon.

Thus the fundamental limitations on spectral response are these internal response limitations: the long wavelength-limiting response set by the thickness of the silicon chip, and the short wavelength limit set by the onset of photoemission into the oxide. (Reflection losses constitute another limitation; these will be discussed as they relate to specific types of devices.)

2. Modulation Transfer Function

The modulation transfer function (MTF) describes the fidelity of an imaging system by giving the attenuation of an image signal as a function

of spatial frequency. In CCD image sensors, the MTF is the result of several parameters: the photoelement aperture, the optical cross talk, the charge-transfer inefficiency (CTI), and the preamplifier characteristics. The resultant MTF is, according to the fundamentals of MTF analysis, equal to the product of the several MTF factors. Thus, one can write

$$(\text{MTF})_{\text{device}} = (\text{MTF})_{\text{aperture}}(\text{MTF})_{\text{cross talk}}(\text{MTF})_{\text{CTI}}(\text{MTF})_{\text{preamp}}.$$

To illustrate how the several relevant device parameters determine the resultant MTF of the device, consider the MTF factor for the aperture of a representative line-scan image sensor, the Fairchild CCD122, along the length of the device. The photoelements are on a $13\text{-}\mu\text{m}$ pitch. Because of the neutral field region between the photodiodes, the lateral response profile of an element is approximately trapezoidal in shape, as shown in Fig. 5, with base dimensions of approximately 8 and $18\text{ }\mu\text{m}$. The MTF for such a profile is given by

$$(\text{MTF})_{\text{aperture}} = \left(\sin\left(\frac{\pi}{2} \times \frac{f}{f_N}\right) / \left(\frac{\pi}{2} \times \frac{f}{f_N}\right) \right) \times \left(\sin\left(\frac{5}{13} \times \frac{\pi}{2} \times \frac{f}{f_N}\right) / \left(\frac{5}{13} \times \frac{\pi}{2} \times \frac{f}{f_N}\right) \right) = 0.493,$$

where f_N is the Nyquist spatial frequency, that is, the frequency where there is one image line-pair for every two photoelements.

The MTF factor for cross talk will be discussed in the following section. The MTF factor for charge-transfer inefficiency in the readout register is given by one of the following two equations. For a single readout register (and operating in the noninterlaced mode), it would be

$$(\text{MTF})_{\text{CTI}} = \exp[-N\epsilon(1 - \cos \pi f/f_N)],$$

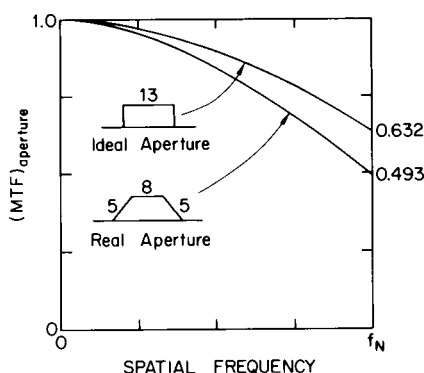


Fig. 5. The MTF factor for the aperture of a line-scan image sensor compared to that for an ideal square response profile. The photoelements are on a $13\text{-}\mu\text{m}$ pitch and the channel stop regions shared by adjacent photoelements are $5\text{ }\mu\text{m}$ wide.

where N is the number of transfers from the particular photoelement to the preamplifier and ϵ the CTI per transfer. For the typical line-scan imager with two readout registers, this becomes

$$(\text{MTF})_{\text{CTI}} = \exp[-N\epsilon(1 - \cos 2\pi f/f_N)].$$

From this expression it may be seen that there is no CTI degradation at the Nyquist frequency but, instead, the largest effect is at half the Nyquist frequency. In typical devices in the Fairchild buried-channel CCD product family, this MTF factor is larger than 0.90 over the full specified frequency range.

3. Optical Cross Talk

The term “optical cross talk” is used here to refer to the phenomenon whereby signal charge generated by light incident on one photoelement is collected at some neighboring photoelement. It may be caused either by optical scattering or by diffusion of photoelectrons that have been excited by penetrating radiation in the deep neutral silicon region. Optical scattering is typically only of the order of 3% in magnitude and is, therefore, generally negligible. The second effect may be larger than this but only in the infrared portion of the spectral response; it will be described next.

Consider an infinitesimal diameter beam of collimated monochromatic radiation falling at normal incidence onto a large area imaging device. Photoelectrons generated in the lower neutral region will, on the average, diffuse laterally a distance comparable with the depth of generation in the neutral region; at least this will be the case if the diffusion length is relatively long. Those photoelectrons that reach the depletion layer will be

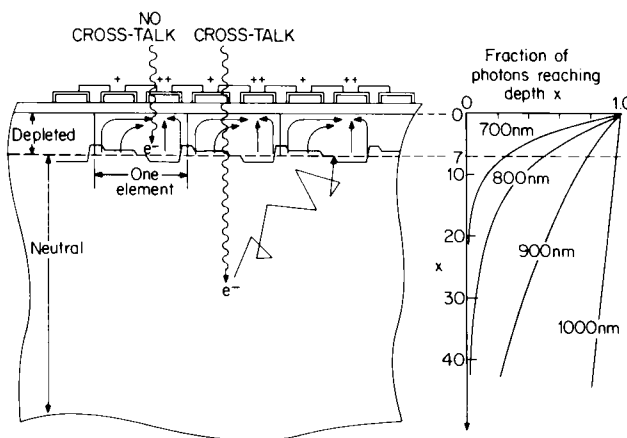


Fig. 6. The optical cross-talk process [21].

collected at a nearby potential minimum for electrons, being guided by local electric fields. Figure 6 depicts this process; it is the basis of the cross-talk model. The model neglects any reduction of the depletion depth or absence of depletion at the channel stops between elements. The assumption of normal incidence is generally a good assumption because even with small f /number optics the high refractive index of silicon (>3.5) results in substantial internal collimation normal to the surface.

The most generally useful form for the characterization of this cross talk is the MTF factor, as previously introduced. This is computed by integrating the signal-spreading process just described over a large sine-wave-modulated area of irradiation. The mathematical treatment of this problem has been described by Seib [19].

An example of a family of calculated MTF curves is shown in Fig. 7 for the case of $L_D = 10 \mu\text{m}$ and $L_n = 100 \mu\text{m}$. From this example, one can see the most prominent feature of the optical cross talk, characteristic of most silicon imaging devices, namely, a large decrease in MTF as the wavelength increases from approximately 800 to approximately 900 nm. This large decrease occurs at spatial frequencies as low as approximately 10 line pairs/mm. (Because the absorption coefficient increases slightly with temperature, these curves should be corrected for large departures from room temperature; see Table I.) Since L_n may vary significantly

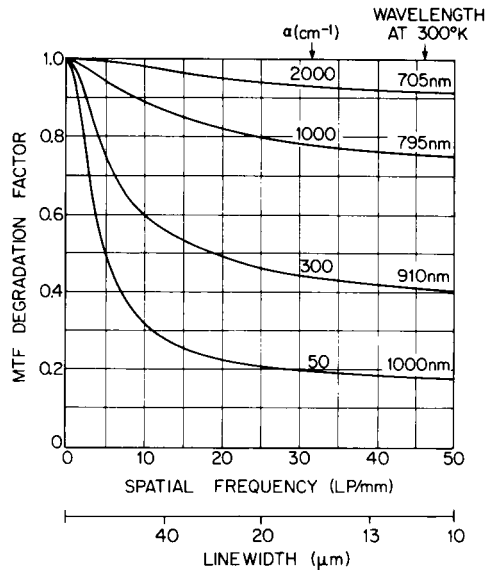


Fig. 7. Calculated MTF degradation factor versus spatial frequency and wavelength for the case of $L_D = 10 \mu\text{m}$ and $L_n = 100 \mu\text{m}$ [21].

TABLE I
Absorption-Length Data for Silicon at Two Temperatures

α^{-1} (μm)	λ (nm)		α^{-1} (μm)	λ (nm)	
	300 K	77 K		300 K	77 K
2.5	600	540	50	940	870
5	705	605	200	1000	940
10	795	715	1000	1060	980
25	880	815			

between devices, from approximately 10 to approximately 100 μm , Fig. 7 may be regarded as a nearly worst case.

Thus a limitation on performance is cross talk at the longer wavelengths. It is not a fundamental limitation because, in principle, a deeper depletion region could be incorporated or an additional junction could be added below, but the values presented here are characteristic of devices that are commercially available today.

4. Dark Signal

Dark signal, also referred to as dark signature, fixed pattern noise in the dark, etc., can include a number of phenomena. Some of these are (1) dark current averaged over the full array; (2) dark-current shading effects (long-range effects across a device); (3) dark-current short-range effects; and, in some designs, (4) MOSFET threshold voltage variations. Although dark current has fundamental characteristics, its practical lower limits in various imaging device structures are not set by fundamentals at all but rather by impurity levels and metallurgical defect levels. Only a few general statements on this subject are appropriate here.

In the majority of cases, near room temperature or lower, and in the absence of high electric fields within the silicon, dark current is determined by depletion volume and depletion area at the surface. The basic relationship is

$$j_D = \frac{qn_i}{2} \left(\frac{L_D}{\tau_0} + S_0 \right),$$

where j_D is the dark-current density, n_i the intrinsic carrier concentration, L_D the depletion length, τ_0 the bulk dark-current parameter, and S_0 the surface dark-current parameter. In many cases the parameters τ_0 and S_0 have only small temperature dependencies so that the dark current varies approximately as n_i . This is a precisely known parameter in silicon and is

TABLE II
Typical Temperature Dependence of Dark Current in Silicon Imagers

Temperature (°C)	n_i (cm ⁻³)	Relative dark current	Approximate temp- erature dependence
55	1.2×10^{11}	10.0	Doubles every 10°C
25	1.2×10^{10}	1.0	Doubles every 8°C
0	1.2×10^9	0.10	Doubles every 7°C
-22	1.2×10^8	0.010	Doubles every 6°C
-40	1.2×10^7	0.0010	Doubles every 5°C

what leads to the convenient ways to estimate temperature dependence shown in Table II.

In CCD imagers that have photodiodes for the sense elements, the major portion of the surface area may not be depleted. In that area, the dark-current density can be significantly less, especially if the depletion length is made to be small. Similarly, in a novel CCD design called "virtual phase" [20], part of the register itself can be covered with an undepleted *p*-type layer (over a buried *n*-channel), which again eliminates the surface component of dark current from at least that portion of the register. The cited reference on virtual phase CCD design reports a CCD image sensor with dark-current densities at 25°C as low as 0.4 nA/cm². The device is a frame-transfer area imager.

B. Line-Scan Image Sensors

1. Designs

A basic line-scan image sensor of the type that is commercially available today consists of a row of photoelements and two CCD readout registers. The photoelements may be either of the CCD type with poly-Si covering them or they may be of the floating-source type, i.e., the photodiode type, without any gate material covering the sensor area.

A preferred type of design for the readout registers is the two-phase, buried *n*-channel structure using two levels of poly-Si. The device is essentially the same as described by Kim [5] in 1974. Transfer gates separate a single photogate from the pair of readout registers. The registers are coupled to a single gated-charge-integrator preamplifier such that one pair of clocks drives the two registers; the preamplifier reset clock runs at twice the frequency of the transport clocks. The buried *n*-channel design is preferred for reasons of speed and high CTE over a wide dynamic range. The CCD type of photoelement is preferred where the most precise

charge readout from the device is important. The photodiode-type of photoelement is preferred where higher quantum efficiency over the full visible spectral range is important.

In some of the designs, a second layer of metallization is used to minimize unintentional photosignals collected in the registers.

2. Fabrication

An n -channel CCD line-scan image sensor can be fabricated much like a high-density NMOS memory device. They both use two levels of polysilicon gates. Ion implantation is essential as a means of doping and is used at least twice in the course of making a typical two-phase device: once to form the n -channel and again to form the barriers that are located under the gaps in the first level of poly-Si along the CCD registers [5].

Before the silicon can be fabricated, however, a set of masks is required, and because line-scan image sensors are typically close to an inch in length, the conventional method of the past, namely, of generating a master from a reticle by 10:1 reduction, is not practical per se. In order to make a master pattern of such length by this method, it is necessary to "compose" the pattern by reductions from two or more reticles. In some cases, the "composition" requires at least one repeat of a central block of the device structure. The accuracy required for this composing is quite impressive, yet it has been done successfully in production. For example, for an array with photoelements on 13- μm centers and with a requirement for a uniformity of response on the order of 5% or better, the alignment at the butt joints must be controlled to better than 1 μm . This requires not only extreme control in stepping position, but a corresponding control in the precision of the 10:1 optical reduction.

With the advent of electron-beam mask making, the optical composing method just described is no longer required. However, the extremely precise control of the master plate position during electron-beam writing is, of course, still essential.

In the processing of the silicon wafer, special attention, in the case of imaging devices, is generally given to parameters that affect junction leakage current, i.e., device "dark current," and to the optimization of photoresponsivity and its uniformity. Bulk gettering and surface-state annealing work together to favorably influence all of these performance parameters. Hence a great deal of attention is generally given to achieve the best possible gettering and annealing processes. As discussed earlier, dark-current densities as low as 0.4 nA/cm² at 25°C have been reported [20]. With such low dark current densities, the minority carrier diffusion length in the p -type bulk is often 50–100 μm or longer. From the principles outlined in Section II.A, it may be seen that such long diffusion

TABLE III
Some Line-Scan Image Sensors Commercially Available or in Development^a

Device type ^b	Product code	Number of elements	Element size ^c (μm)	Active length (mm)	Spectral response type	Maximum sample rate (MHz)	Saturation charge (electrons)	Dynamic ^d range (typical)	Signal level at low end of DR (electrons)	Responsivity ^e (typical) (V/μJ cm ⁻²)
<i>Fairchild</i>										
PD/CCD	CCD111	256	13 × 17	3.3	1	10	1.1 × 10 ⁴	2500:1	500	1.2
PD/CCD	CCD133	1024	13 × 13	13.3	2	20 ^f	1.4 × 10 ⁶	2500:1	600	3.0
CCD	CCD121H	1728	13 × 17	22.5	1	10	1.2 × 10 ⁶	2500:1	500	1.0
PD/CCD	CCD122	1728	13 × 13	22.5	2	2	1.3 × 10 ⁶	2500:1	500	3.0
PD/CCD	CCD142	2048	13 × 13	26.6	2	2	1.3 × 10 ⁶	2500:1	500	3.0
PD/CCD	CCD143	2048	13 × 13	26.6	2	20 ^f	1.4 × 10 ⁶	2500:1	600	3.0
<i>Reticon</i>										
PD/MOS	Series C	256, 512, 1024	25.4 × 25.4, 432	≤26.0		10	2.5 × 10 ⁷	1500:1 at 1.5 MHz ^h		
PD/MOS	Series G	128, 256, 512, 1024	25 × 26	≤25.6		1	2.0 × 10 ⁷	1500:1 at 1 MHz ⁱ		
PD/MOS	Series H	1024, 1728, 2048	15 × 16, 300	≤30.7		3	1.1 × 10 ⁷	3000:1 at 1 MHz ^j		
PD/MOS	Series B	512, 1024	25.4 × 25.4, 610	≤26.0		40 ^g				
PD/MOS	Series F	936, 1872	15 × 16, 750	≤28.1		20 ^g				
PD/CCD	CCPD	256, 1024	16 × 16	≤16.4		2	9.0 × 10 ⁷	500:1		
PD/MOS	Series S	128, 512, 1024	25 × 2500	≤25.6				10 ⁴ :1 at 300 kHz ^k		
(Spectrometric)								10 ⁵ :1 at 200 kHz		

Other Manufacturers¹

PD/CCD(N)	1024	14 × 14	14.3	
CCD(N)	2048	14 × 14	28.7	3
μPD792D	1024	15 × 15	15.4	2
PD/CCD(T)	1728	15 × 15	25.9	≥0.5
PD/CCD(T)	1024	25	25.6	1
PD/MOS(O)	1728	15	25.9	1
CCD(O)	2048	12	24.6	
PD/MOS(H)	1024	13	13.3	2
PD/CCD(H)	1024	13	13.3	
CCD(H)	2048			4
PD/MOS(M)	1024	28	28.7	
PD/CCD(M)	1024			1 ^m
MN8027	2048	14 × 14	28.7	
PD/CCD(M)	1024	13	13.3	2
CCD(F)				1
PD/CCD(S)	2048	14 × 14	28.7	

^a This table is based on the best available information.

^b PD = photodiode sense elements; MOS = MOS scan generators.

^c First dimension is also the center-to-center spacing unless noted otherwise.

^d Dynamic range is defined as saturation signal divided by RMS temporal noise.

^e Standard tungsten lamp at $T_c = 2854$ K filtered by Schott BG-38 filter to give a source spectrum with a range of 400 to 700 nm (20% limits).

^f Dual output: 10 MHz per output.

^g Four parallel outputs.

^h Obtained with Evaluation Ckt #RC100B.

ⁱ Obtained with Evaluation Ckt #RC1024G.

^j Obtained with Evaluation Ckt #RC1728H/LN.

^k Obtained with Evaluation Ckt #RC1024SA.

^l (N) = NEC; (T) = Toshiba; (O) = OKI; (H) = Hitachi; (M) = Matsushita; (F) = Fujitsu; (S) = Sharp.

^m Dual output: 0.5 MHz per output.

lengths give a high internal responsivity to wavelengths as long as 900 or 1000 nm. To achieve a uniform response at these wavelengths, it is necessary to produce uniform diffusion lengths across the entire length of these arrays. This is a particularly difficult parameter to control, especially when the wafer contains any "swirls" of precipitated impurities that may often cause lifetime fluctuations in their vicinities.

3. Performance of Representative Devices

At present, there are several long line-scan image sensors commercially available. The most widely used class of device has approximately 1700 photoelements and is primarily used for optical scanning of the $8\frac{1}{2}$ -in. direction of an $8\frac{1}{2} \times 11$ -in. page, giving a resolution of 200 lines/in. Another similar class of device has approximately 2000 elements. Table III shows some of the salient properties of these devices. The table includes devices scanned by MOS shift registers; 19 CCD- and 21 MOS-scanned device types are listed. Note that these are some of the longest VLSI structures, if not actually the longest, manufactured today. Several listed in the table are very close to 25 mm in length and a few are longer. The photoelement size ranges from 12 μm to 28 μm .

The size of the photoelement in the devices with the largest number of elements is primarily dictated by layout design rules rather than by optical requirements. For example, devices with 13- μm photoelements employ a multilayer polysilicon CCD gate structure in which the spaces are approximately 4 μm wide; this was the design rule when the first device of the family was developed.

Table III also shows that some of these devices are very fast, with maximum sampling rates on the order of 20 MHz. These upper limits are typically set by the design of the on-chip preamplifier. Because faster preamplifiers dissipate significantly more power, it is advantageous to limit the speed to that which is most often adequate and thus to hold down the power dissipation. For example, the CCD122 shown in the table operates to 2 MHz and dissipates 180 mW while the CCD133 operates at 20 MHz and dissipates 270 mW.

The dynamic range is typically well over 1000:1. This range extends from an upper limit of charge-storage capacity of the photoelement, or of the register, down to the random noise of the preamplifier. (Dark signal in these devices typically determines a lower dynamic range near room temperature, the level depending on the integration time.)

A compilation of the saturation parameters for several Fairchild image-sensor products is given in Table IV. The line-scan image sensors

TABLE IV
Saturation Exposure Levels of Fairchild CCD Image Sensors^a

Device type:	CCD111	CCD121H	CCD122	CCD133	CCD211
Spectral type:	I	I	II	II	III
Saturation Charge Level:	1.0×10^6	1.2×10^6	1.3×10^6	1.4×10^6	3×10^6
Type of source spectrum					
2854 K, unfiltered ^b					
Photometric	0.25	0.35	0.25	0.35	0.10
Radiometric	1.2	1.5	1.2	1.6	0.50
2854 K + F900 filter ^c					
Photometric	0.60	0.80	0.60	0.80	0.30
Radiometric	0.40	0.55	0.40	0.55	0.20
2854 K + F700 filter ^d					
Photometric	1.5	2.0	1.5	2.0	0.80
Radiometric	0.50	0.67	0.50	0.67	0.28
Monochromatic (600 nm)	0.42	0.55	0.42	0.60	0.23
					Units (el/pixel)

^a Levels are based on the "typical" values given in the respective data sheets, except for the CCD121H. The basis values are in italics.

^b $1 \mu\text{J}/\text{cm}^2 = 0.22 \text{ lux sec}$.

^c $1 \mu\text{J}/\text{cm}^2 = 1.5 \text{ lux sec}$. The F900 filter blocks wavelengths above approximately 900 nm. It is a Corning 1-75 glass filter 3 mm thick.

^d $1 \mu\text{J}/\text{cm}^2 = 3.0 \text{ lux sec}$. The F700 filter blocks wavelengths above approximately 700 nm. It is a Schott BG-38 glass filter 2 mm thick.

have a saturation charge of approximately 10^6 electrons. The preamplifier noise is typically approximately 300 electrons rms. Most of this noise is reset noise (or kTC noise) associated with the resetting of the input node voltage of the preamplifier; by correlated double sampling, this noise component can be eliminated completely, thus providing a still wider dynamic range.

a. Photodiode Sense Elements versus CCD-Type Sense Elements. As previously described, there are two fundamentally different kinds of sense elements in common use today in line-scan image sensors: CCD-type structures with semitransparent gates and photodiodes without them. The first type uses the gate to control the potential of the sense element, whereas the second uses the conductive undepleted portion of an n -type layer for this purpose. The second type has the clear advantage of higher photoresponse due to the absence of the semitransparent gate, which, in general, causes reflection loss and some absorption loss. Figure 8 shows spectral response curves for these two types; type I in the figure is the CCD-type and type II is the photodiode type. (The third curve will be discussed later.) These curves are examples based on the Fairchild product line. The curves are smoothed so that the interference maxima and minima of the actual spectra are eliminated. Examples of the actual spectra are shown in Fig. 9. Because of the peculiarities of the actual spectra, the smoothed spectra are more reliable for estimating performance of typical devices. In Fig. 8 it is seen that the semitransparent gate in the type II devices (doped polysilicon) causes a decrease in response of

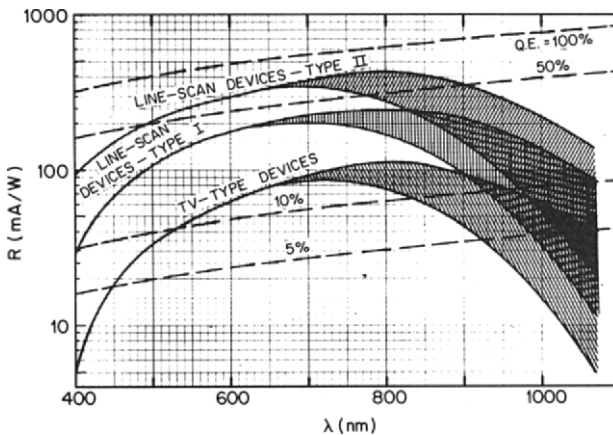


Fig. 8. Typical internal spectral response, smoothed to eliminate interference effects, for Fairchild CCD image sensors. (These data are valid for the finished device with the uncoated window in place.)

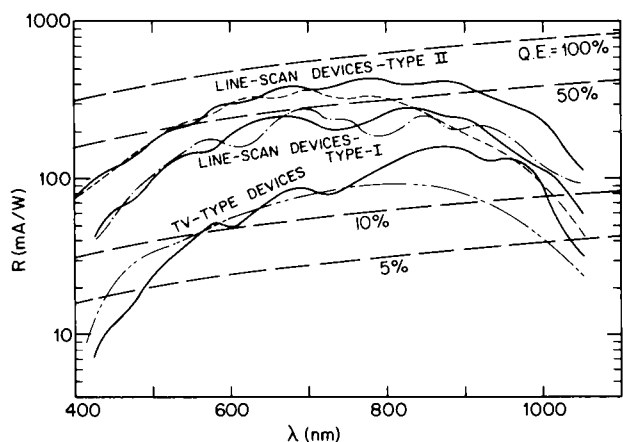


Fig. 9. Typical (unsmoothed) internal response curves.

approximately 35% over most of the spectral range. The diverging curves at longer wavelengths indicate the range of variation in response due to diffusion-length variation from one device to another (see the discussion of the principles in Section II.A).

In many applications of these devices, broadband illumination is used. The responsivities of these devices to a few broadband sources are given in Table V. Monochromatic response at 600 nm is included in the table as a convenient reference. These responsivities may be converted into photometric units by the use of the conversion factors given in the table footnotes. The spectra of these broadband sources are shown in Fig. 10. They

TABLE V
Typical Internal Responsivities of Fairchild CCD Image Sensors

Type of radiation	Responsivity (mA/W)		
	Line-scan devices		TV-type devices
	Type I	Type II	
2854 K, unfiltered ^a	~60	~100	~25
2854 K + F900 filter ^b	170	300	60
2854 K + F700 filter ^c	140	250	45
Monochromatic (600 nm)	180	300	60

^a 1 $\mu\text{J}/\text{cm}^2 = 0.22 \text{ lux sec.}$
^b 1 $\mu\text{J}/\text{cm}^2 = 1.5 \text{ lux sec.}$ The F900 filter blocks wavelengths above approximately 900 nm. It is a Corning 1-75 glass filter 3 mm thick.
^c 1 $\mu\text{J}/\text{cm}^2 = 3.0 \text{ lux sec.}$ The F700 filter blocks wavelengths above approximately 700 nm. It is a Schott BG-38 glass filter 2 mm thick.

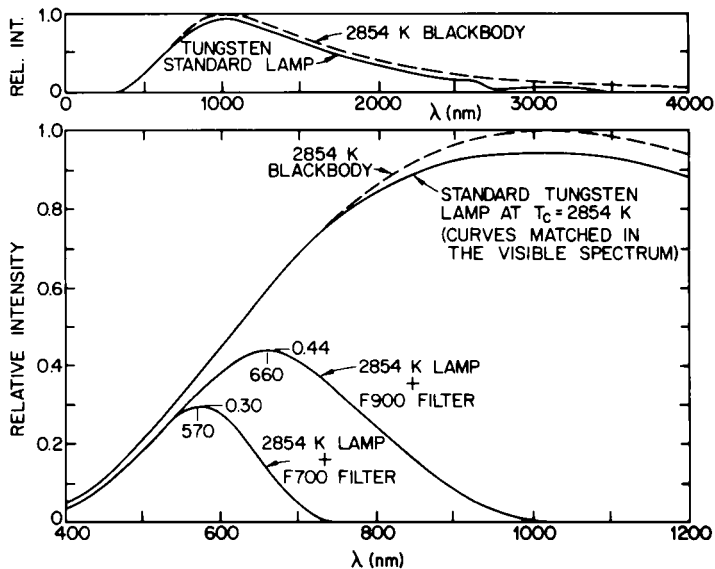


Fig. 10. Spectra of three broadband light sources.

are normalized such that the peak of the spectrum for a blackbody of the color temperature and luminosity of the tungsten lamp is unity.

b. MTF and Optical Cross-Talk Effects. In Section II.A, the fundamentals of optical cross talk and the resulting MTF degradation factors were discussed. Here an example will be given of a representative set of data for a line-scan image sensor.

The first data to be presented are the MTF data (see Fig. 11). The sine-wave-modulated radiation pattern that is required to make the measurements was approximated by the use of a chrome bar-pattern mask and a reflective microscope objective. The reflective objective was out of its best adjustment to such a degree that at 25 lp/mm it gave approximately a sine wave image at close to 100% amplitude. It is noted that for wavelengths of 800 nm and less the MTF typically remains above 0.40; this is regarded as very high MTF.

Another representation of optical cross talk may be obtained by focusing a spot of light smaller than one photoelement on the center of one element and reading the signal at neighboring elements. A typical result is shown in Fig. 12. [21]. It may be noted that at 900 nm the second and third nearest neighbors are showing significant pickup.

A second optical cross-talk effect in line-scan imagers is direct charge pickup in the registers, an effect that causes a shift in the black level when

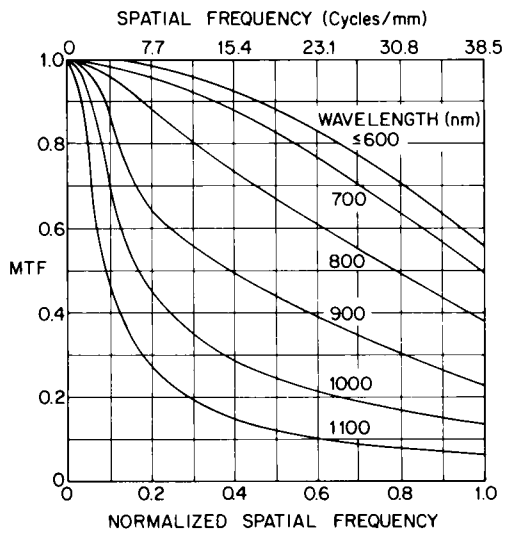


Fig. 11. Typical in-phase MTF of the Fairchild CCD121H line-scan image sensor versus wavelength and spatial frequency. The Nyquist frequency is 38 line pairs/mm [21].

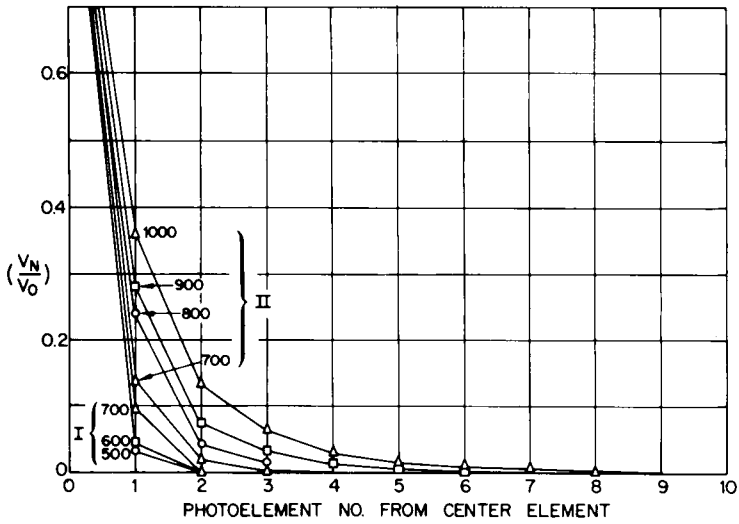


Fig. 12. Typical optical cross talk of the Fairchild CCD121H line-scan image sensor characterized using a light spot. The photoelement spacing is 13 μ m and the wavelengths are in nanometers. Group I data: 5- μ m spot, 20 $\times$ Nikon objective lens; group II data: 10- μ m spot, 15 $\times$ Beck reflective objective (not corrected) [21].

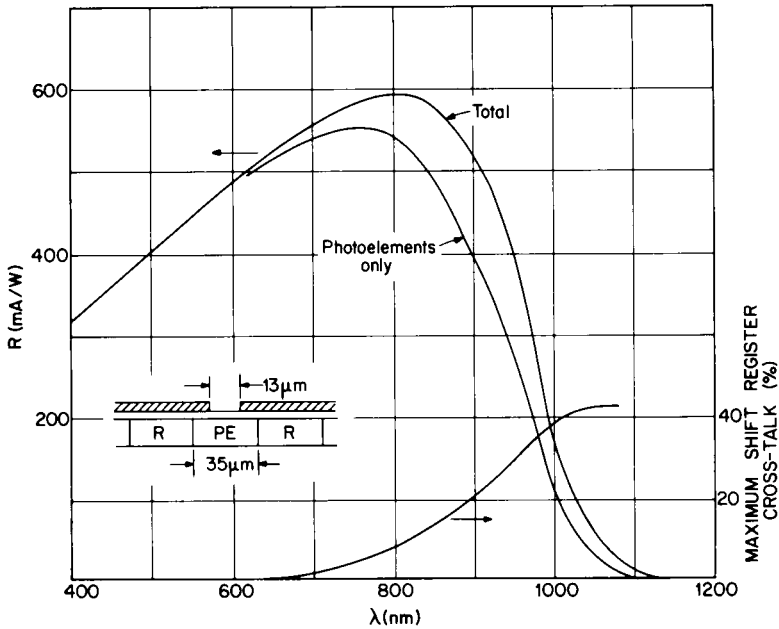


Fig. 13. Theoretical internal spectral response for the CCD133 line-scan image sensor showing the amount of cross talk in the registers with $L_D = 7 \mu\text{m}$ and $L_n = 50 \mu\text{m}$ [21].

most of the array is illuminated. Figure 13 [21] shows this effect as a function of wavelength as estimated with the aid of the previous cross-talk model. The estimate agrees well with measurements on representative sample devices.

C. Area Imaging Devices for Television

1. Architecture

An area image sensor that is to be used for conventional television must be capable of being scanned at 50 or 60 fields/sec, interlaced, and must not have an objectionable amount of image smear associated with the scanning structure or the scanning process. The CCD image sensors that have been proposed for this application have four distinctly different architectures: frame transfer (FT), interline transfer (ILT), line transfer (LT), and the type with a MOS photoarray and CCD readout register (MOS/CCD). Schematic diagrams of these are shown in Fig. 14. The frame-transfer type has been attractive for device developers because of its relative simplicity. For example, the RCA "Big SID" device has been

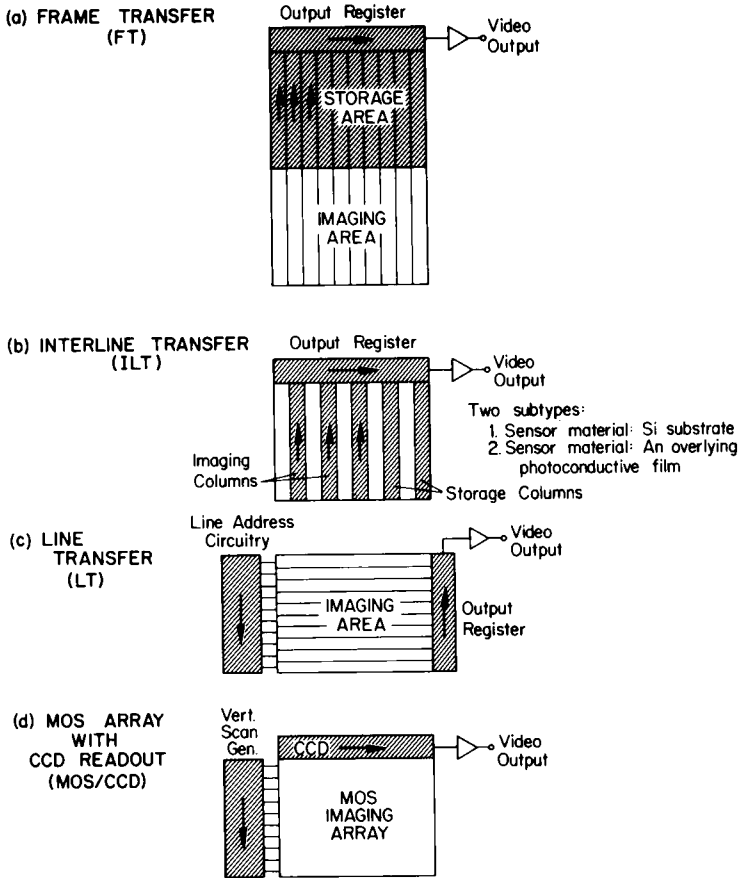


Fig. 14. Four types of CCD imagers for TV.

fabricated [14] using only two masking levels to define the entire imaging and storage areas, namely, a channel stop pattern and a doping pattern for forming sets of three-phase clocks in one continuous sheet of polysilicon. The interline transfer type is considerably more complex to fabricate but has the advantage over the FT type that it results in a smaller die area. For example, the image-sensor size may be determined by a requirement to use a standard TV format size or by design rules that dictate how short the output register can be. In each of these cases, the width of the device is fixed. Clearly, then, the height of the ILT-type device will be significantly less, leading to a significantly smaller chip. Of course, chip size and process complexity are major factors in the manufacturing cost of a device, making this trade-off between these two types of devices of key importance.

TABLE VI

Manufacturer	Device type	Product code (or ref.) ^b	Number of elements (V × H)	Size of raster (diagonal) (mm)
Fairchild	ILT	CCD211	244 × 190	7.2
	ILT	CCD221	488 × 380	14.4
RCA	FT	SID52501	512 × 320	12.2
Toshiba	FT	TAS-III	512 × 340	10.5
Matsushita	MOS/CCD	(R1)	256 × 404	11.0
	FT	(Product)	512 × 404	10.7
	PC/BB/CCD ^c	(R2)	506 × 413	18.6
Sony	ILT	(Camera)	492 × 226	10.3
	ILT	(R3)	492 × 245	11.0
Hitachi	MOS	(R4)	484 × 384	11.0
T.I.	FT	(R5)	490 × 328	9.8
NEC	ILT	(R6)	490 × 384	16.0

^a This table is based on the best available information.

^b (R1) = Terakawa, May 1980 [23]; (R2) = Terui *et al.*, ISSCC 1980 [24]; (R3) = Nikkei Electronics, 1980, 3.3, pp. 69–74 [25]; (R4) = M. Aoki *et al.*, ISSCC 1980 [26]; (R5) = Hynecek, IEBM 1979 [20]; (R6) = Tshihara, ISSCC 1980 [27].

^c Photoconductive film scanned vertically by bucket-brigades and horizontally by a CCD register.

^d Blooming suppression achieved by operating much below saturation (= 10×).

The LT-type device has received relatively little attention in its development. In 1972, an RCA team reported on successful fabrication and operation of a 32×44 array [22]. Since this number of elements is so far short of that needed for television, it has been omitted from Table VI, which will be discussed later.

The MOS/CCD-type array has been shown in a Matsushita letter [23] to give very satisfactory performance. As is discussed in that letter, it has previously not been possible to achieve satisfactory signal-charge transfer out of the long vertical drain lines. The letter describes a method for accomplishing this that requires the use of an additional transfer gate and a floating diffusion whose potential can be altered by an additional capacitive element. The process is called "charge priming transfer."

Table VI describes some of the arrays that have been reported to be under development or that are commercially available. These devices clearly all fall into the VLSI category in view of their chip sizes that range from 6.1×6.3 up to 15×20 mm. (Texas Instruments's 800×800 -pixel device, which is not listed here because it is not a television-type device,

Large Area Image Sensors for Television^a

Chip size (mm)	Blooming suppres- sion	Indicated use	Gate material over photoelements	SNR in color camera at $f/2$ (dB)
6.1 × 6.3	No	General	Poly-Si	
10.1 × 12.1	No	General/NTSC	Poly-Si	
12.7 × 19.0	No	General/NTSC	Poly-Si	
10.0 × 14.5	No	Two-chip color cam	Poly-Si	
	Yes		None	
10.3 × 13.6		General/NTSC	Poly-Si	
	N/A	One-chip color cam	SnO ₂	43 at 500 lux
9.1 × 10.3	Yes	Three-chip color cam	SnO ₂	40 at 700 lux
9.1 × 10.6	Yes	Two-chip color cam	Thin poly-Si	>43 at 500 lux
8.5 × 10.0	Yes	One-chip color cam	None	40 at 500 lux
	No	General/NTSC	Poly-Si	
11.0 × 14.4	Yes ^d	One-chip color cam	None	46 at 600 lux (green channel)

is only slightly bigger than the largest of these in chip area, even though it has significantly more pixels.)

Most of these devices have been designed to have approximately full (monochrome) television resolution. Most also have raster sizes close to that of the $\frac{3}{4}$ -in. vidicon, that is, they have a diagonal length of 11 mm. The number of elements in the horizontal direction is in most cases close to either 320 or 380. The Sony devices for two-chip and three-chip color cameras are exceptions with significantly fewer elements; these designs achieve adequate horizontal resolution in their respective cameras by the ways that the devices are aligned to each other.

2. Front-Side versus Back-Side Illumination and Quantum Efficiency

The technology for fabricating a solid-state array that can receive its optical input from the back side was developed in the late 1960s as part of the silicon-vidicon development. The advantage of back-side illumination is higher quantum efficiency. To achieve the proper performance over the visible spectrum, it is necessary to thin the silicon to the order of 10 μm . However, the majority of devices being developed in recent years employ front-side illumination. Unit fabrication cost is a very large consideration. Another consideration is the development of the means for achieving comparable front-side quantum efficiencies (see Section II.C.4.e).

3. Fabrication

A VLSI CCD imager can be made by much the same wafer-processing technology that is used to make other types of integrated circuits. One applicable process is the buried n -channel process using polysilicon gates. The imager process may use either two or three layers of polysilicon. It may also use silicon nitride. Because silicon nitride has a refractive index well above that of SiO_2 (2.0 as compared to 1.47), it can be used to reduce reflection losses at polysilicon surfaces and thus improve quantum efficiency.

In addition to these aspects of device fabrication, however, there is one phenomenon that degrades an imager significantly but might not hurt a purely electronic device at all. This is the effect of small-dimension non-uniformities that can result from mask composing. In an imager, if one photoelement or one row of photoelements is even a fraction of a micrometer larger or smaller than its neighbors, it can have a significantly different responsivity, which can, in turn, result in a visible effect on the image produced. For example, a $0.5\text{-}\mu\text{m}$ change in the $13\text{-}\mu\text{m}$ width of some photoelements of the Fairchild CCD221 (see Table VI) could produce a 4% change in responsivity—a readily visible change.

Of course, there is also the requirement for the very best quality processing relative to particulates. These not only can cause electronic failures but optical blemishes as well.

4. Performance

a. MTF and Aliasing. A CCD imager with a high charge-transfer efficiency can have an MTF that remains high (>0.6) for all frequencies up to the Nyquist limit of the device. Although this high MTF reproduces image information with high precision (with a lens of comparably high MTF) facilitating detection of fine detail, it results in a high-amplitude beating phenomenon between high spatial frequencies of the image and the spatial frequency of the photoelement spacing. This beating is known as *Moiré*; under certain conditions it can be readily observed from frequencies above Nyquist frequency to well under half of that frequency. The phenomenon has been thoroughly described elsewhere (see for example, Sequin and Tompsett [3]).

From a basic rule of sampling theory, there is an optimum condition where there are three samples per cycle of signal at the maximum signal frequency. For NTSC (National Television Standards Committee) conditions, where the maximum signal frequency corresponds to 4.2 MHz, or 220 cycles/picture width, this suggests that an optimum number of photoelements per row is 660. However, to date, almost all arrays are using a

number near 380, which gives approximately two samples per maximum signal frequency. Given arrays with this number of photoelements, the camera designer has a choice: either the full frequency range can be detected and the aliasing tolerated or the aliasing may be eliminated completely by the use of an optical low-pass filter. In the first case, very crisp reproduction of detail is possible; see, for example, the image in Fig. 15 produced with the Fairchild CCD221 [28]. In the latter case, a sacrifice in MTF at high frequencies is a necessary result. (One type of typical low-pass filter would be an optical element that produces a pair of images from a single image, the separation of the images being half of a cycle at the Nyquist frequency. One example is a block of a birefringent crystal such as quartz.)

b. Readout Image Degradation. All of the preceding CCD imager architectures give rise to some degree of image smearing while the signal charge from one portion of an image is transferred past other portions of the image. This is even true of the MOS/CCD-type device. The smearing mechanism and the amplitude of the smear in current designs of TV imagers will now be discussed.

In the case of frame-transfer imagers, there is a vertical smear compo-



Fig. 15. Image produced with the Fairchild 488×380 -element interline transfer image sensor.

nent of the signal that is equal to the average signal in any one-column register divided by the duty factor of the signal-transfer process, that is, the signal-transfer time divided by the time that the clocks for the photo-sensing area are stopped. For example, consider the transfer process indicated in the tentative data sheet for the Toshiba TAS-III 512×340 -element device (a frame transfer imager) [29]. The transfer clock frequency is given as 500 kHz while the line shift frequency is 15.75 kHz; the ratio of these frequencies is 32. Therefore, $\frac{1}{32}$, or 3.1%, of the optical input signal is smeared vertically.

In ILT imagers, there are opaque vertical registers interdigitated with columns of sensor elements; in this type of imager it is optical cross talk into these registers that leads to image smearing in the vertical direction [21]. For example, cross talk from a very bright spot in the image will, because of the way the device is clocked, result in a uniform increase in signal in all the charge packets moving through each shift register adjacent to illuminated photoelements. At the display, this increase in signal appears as a vertical smear, which is uniform over the full height of the picture. Figure 16 shows a schematic cross section of the device. The phenomenon is the same as has been previously described for line-scan imagers, but whereas in the line-scan device the metal extends from the opaque registers over the photoelements by $11 \mu\text{m}$, in ILT-type imagers it is necessary to employ much smaller dimensions for this overlap. In this discussion, we shall consider designs with overlap dimensions of 2 to 6 μm . The reason for holding this dimension small is that it directly affects

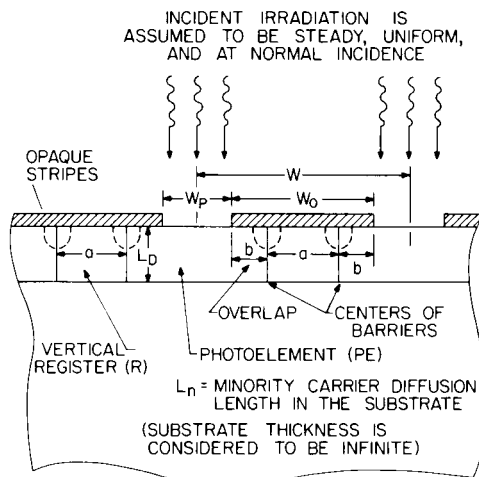


Fig. 16. Model for the calculation of deep-carrier cross talk in an ILT image sensor [21].

the responsivity of the device and/or the cell size. In the Fairchild CCD211, which is a 244×190 -element ILT image sensor with a horizontal cell dimension of $30 \mu\text{m}$ and a horizontal aperture of $14 \mu\text{m}$, an increase of overlap from 2 to $6 \mu\text{m}$ would result in a decrease in responsivity of more than 50%. Figure 17 shows how the level of the smear signal in the CCD211 varies with wavelength. Here a parameter called the charging ratio (r) is plotted; the inset in the figure shows how the smear signal varies with this parameter and the size of the bright spot causing the smear. The charging ratio is the ratio of the shift-register cross-talk photo-current to the photoelement current.

The predicted curve in Fig. 17 shows that at 600 nm the charging ratio should be less than 1%. However, as shown in the figure, typical devices have a charging ratio, at wavelengths of 600 nm or less, of approximately 2%. This excess is also seen to be more or less independent of wavelength. This excess cross talk is attributed to optical scattering in the polysilicon gate material; the scattering could occur both at oxidized grain boundaries and at sloped edges of any of the exposed polysilicon layers. Thus, for wavelengths less than 650 nm, the two types of devices are comparable in performance, but at longer wavelengths, the interline transfer device tends to have a larger amount of smear.

In the LT type of device, there is a horizontal smear component ex-

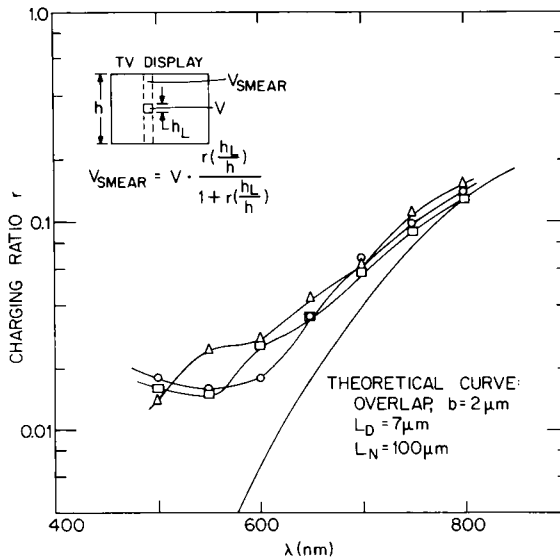


Fig. 17. Charging ratio versus wavelength for three typical CCD211 ILT image sensors [21].

actly analogous to that in the frame-transfer type; however, here the charge-transfer duty factor is not as arbitrary. The duty factor here is approximately determined by the number of registers in the device. For an NTSC-type device with 488 registers, the smear factor would be $(488)^{-1} \cong 0.002$. However, a line transfer device of this size has not been reported.

The MOS-type device has more or less the same vertical optical smear phenomenon as does the ILT type; that is, the vertical drain lines are photosensitive, and therefore, they pick up optical cross-talk signals as do vertical registers in the ILT devices even though they are 100% opaqued. The difference lies in the geometry of the junction areas of the vertical drain line. (In principle, the drain lines could be used for only a fraction of the time for signal readout and thus the smear signal could be further reduced, but apparently, performance in this mode of operation has not been reported.)

c. Blooming Effects. Silicon CCD image sensors are known for their property of controlling charge packets with a high degree of precision. However, above the saturation level that inevitably occurs somewhere in the device, the charge will overflow along the registers unless some special overflow sinks are provided. This gives rise to directional blooming, which can be quite noticeable and even objectionable at exposure levels just slightly above saturation. Therefore, in applications where saturation can occur, it is desirable to have overflow or antiblooming sinks. One example of a TV imager with such a structure is the one described by Kanoh *et al.* [30] for use in a two-chip color camera. The device has 492×226 elements and a unit cell of $14 \times 36 \mu\text{m}$. Although the $36\text{-}\mu\text{m}$ horizontal dimension of the unit cell may appear to be relatively large, it is significant for its smallness, especially since approximately 25% of that dimension is taken up by the overflow control gate and overflow sink.

However, even with a perfect charge overflow and sinking function, there will still remain any optical cross-talk and charge-transfer smear phenomena. These effects are expected to be the major limitation on future device performance for imagery with very bright highlights.

d. Low Light Level Characteristics. Television-type CCD image sensors have been demonstrated to give detectable images (or bar patterns) at a device illumination level as low as 4×10^{-4} lux (unfiltered 2854 K tungsten) at the standard television frame rate. This excellent low light level performance was achieved even though the peak quantum efficiency was only approximately 20%. This corresponds to a signal level of approximately 60 electrons/pixel for the bar pattern; the signal-to-noise ratio was estimated to be close to 2. This performance was measured on a

developmental version of the Fairchild 488×380 imager previously described at a device temperature of approximately -40°C . (The spectral response was approximately that shown in Fig. 8 for the higher end of the range for TV-type devices.) A key factor in achieving that performance was the incorporation of a high-gain, low-noise preamplifier. The preamplifier in this device is a floating gate amplifier [31] with a minimized input capacitance. At a slower scanning rate, a similar device, the Fairchild 244×190 imager, has been used successfully in an astronomy application down to a noise level of approximately 20 electrons [32]. Several other significant recent developments in the use of low light level, solid-state imagers in astronomy are given in a collection of papers edited by Morgan and McMullan [7].

e. Quantum Efficiency. Additional kinds of integrated circuit technology have been developed for improved internal responsivity (or quantum efficiency) to visible and/or near-infrared radiation. One of these, which is pertinent to television-type image sensors, consists of the use of a polysilicon gate that is much thinner than the conventional 200–400 nm. A thickness of 50 nm was selected so as to give an optimum transmittance in the blue region of the spectrum where the signal-to-noise ratio in a color camera would otherwise be marginal. A large improvement in quantum efficiency is indicated [25].

Another of these new developments is the use of a junction gate for one of the CCD phases in analogy to the junction-gate field-effect transistor. This approach is reported by Hynecek [20]; it is applied to the T.I. frame-transfer imager listed in Table VI. In this design, half of the sensor area has the junction gate and half has a conventional CCD gate.

f. Blemishes. Any discussion of performance of TV-type imagers would be incomplete without some consideration of blemishes. Although some of these imagers have emerged from the development laboratory and are available commercially, there are still open questions regarding what types of blemishes may be acceptable in specific applications. The requirements in one existing application for the Fairchild CCD221 image sensor have been described by Vicars-Harris; this application is in the Air Force Cockpit Television System [33]. The reader who has been concerned with integrated circuit yield but who has not previously considered the yield factors of imagers should be impressed by the number of parameters of importance for each picture element and each stage of the several analog shift registers: (1) dark-signal nonuniformity effects, (2) photoreponse nonuniformity effects, (3) charge-transport defects that may only influence small fractions of a saturation-level signal, (4) the number of random single-pixel blemishes, and so on. There is no doubt that the

development of TV-type imagers with some 200,000 pixels and a chip area of approximately 100 mm² is one of the most ambitious VLSI activities being pursued today.

5. Comparison with MOS and CID Photoarrays

A MOS photoarray consists of photodiodes constituting the floating sources of an array of FETs whose gates are tied together in rows and whose drains are tied together in columns. The drain columns are read or scanned with a MOS shift-register scanner, a CCD output register, or possibly by some other means. The case of a CCD output register has been included in a previous discussion (see Section II.C.1). The case of a MOS shift-register scanner for horizontal readout has not looked like an attractive approach at video sample rates because of the high sense-node capacitance of such a circuit. However, very interesting results have recently been reported by Koike *et al.* [34] on a one-chip color imager using just such a structure. That imager is included in Table VI.

The CID, or charge-injection device, differs from the MOS photoarray in that the photodiodes are biased capacitively and restored in potential once each frame time by capacitively driving the photodiodes into forward-bias injection. The signal may be read as substrate current or may be read nondestructively as displacement current in the column gate lines. As in the case of the MOS photoarray, the most advanced version of the CID has a CCD readout register.

6. Color Imagers

Given a full-resolution black-and-white TV image sensor, it has been straightforward to develop the three-chip color camera using the same type of color separation prism used in the older, tube-type camera. Perhaps the most challenging aspect of the development of such a camera and constituent imaging devices has been the achievement of a satisfactory signal-to-noise ratio in the blue channel where the silicon spectral response tends to be the lowest and where incandescent light sources are the weakest.

In addition to some demonstrations of color imaging with the preceding "full-resolution"-type camera, there has been extensive activity in compromise designs in which various amounts of resolution are sacrificed in order to achieve certain manufacturing cost objectives. These include three-, two-, and one-chip camera designs. Early efforts in this area were reported by Dillon [35], in 1978, in a paper that dealt with color-filter considerations, and by Umemoto [36] and Takamura [37], both also in 1978, in papers that dealt with one-chip color-imager experiments.

In 1978, Toshiba announced a two-chip, NTSC-compatible (two 512×340 -element chips) color camera [38]. The imager chips used in this camera are of the frame-transfer type. One chip senses green and the other senses red and blue in alternating columns.

The previously mentioned array reported by Koike [34] is a NTSC-compatible 488×384 -element imager chip for a one-chip color camera. It is a fully MOS design. An important feature of the design is high blue quantum efficiency. The photodiodes are coated for minimum reflection loss in the blue band and the width of the vertical metal stripes connecting the drains is minimized so as to achieve a net quantum efficiency at 450 nm of up to 50%.

In 1980, Terui *et al.* [24] reported the development of a substantially different type of device for color-imager applications. It utilizes a photoconductive layer deposited onto an XY scanning structure. With this structure, virtually the entire unit cell area is available for sensing, and a net quantum efficiency of 70% at 450 nm has been achieved.

As can be seen from this chapter, a considerable amount of older and newer technology has been applied to the color-imager problem in a very short period of time.

D. TDI and “Snap-Shot” Imaging Devices

1. Principles of TDI Image-Sensor Operation

The time-delay-and-integration (TDI) image-sensor operation consists of moving an image at a uniform velocity across a two-dimensional array of photosensitive elements and clocking the array such that the potential wells for collecting the photoelectrons move in registry with corresponding elements of the image. In this way, the two-dimensional array operates like a linear array but has the advantage of integrating the photoelectronic signal for much longer times. The basic principles of TDI consist of the relationships between MTF and (1) the type of CCD clocking and (2) the average velocity differences between the potential wells and the image. The analysis in any of these cases consists of determining a time-averaged response profile for an individual picture element (a pixel) and then convolving it with optical signals that are sine-wave modulated in the direction of motion. These analyses have been treated by Barbe [39]. The MTF factor at the Nyquist frequency for several types of CCD clocking are as follows:

- (i) two-phase TDI (2 hops/cycle): 0.900,
- (ii) three-phase TDI (3 hops/cycle): 0.955, and
- (iii) four-phase TDI (4 hops/cycle): 0.974.

In comparison, a line-scan imager scanning at a velocity such that the image moves the width of the array aperture in one integration period has a corresponding MTF factor of 0.638. Thus, TDI devices have, in principle, a higher MTF in the image motion direction than do simple line-scan devices, provided that there is no velocity difference. If there is a uniform velocity difference, there will be a second MTF factor. This factor is equal to

$$\frac{\sin[(\pi/2)(f/f_N)(N \Delta v/v)]}{(\pi/2)(f/f_N)(N \Delta v/v)},$$

where f is the spatial frequency, f_N the Nyquist frequency, N the number of lines of TDI integration, and $\Delta v/v$ the fractional velocity mismatch.

2. High-Resolution Design Concepts

High-resolution designs for TDI arrays have some of the same features as CCD memory arrays. This is understandable because in both cases it is desired to achieve the highest possible density of charge packets in a two-dimensional array for a given set of design rules. Perhaps the major difference is that, in the case of the imaging device, the time-averaged spacing of the charge packets should be equal or nearly equal in both directions in the array. In the image motion direction, the highest pixel densities have been achieved with the use of ripple-clock designs. In the other direction, interlaced readout has been used to achieve pixel densities of two and three times the stage density of the readout shift register. An example of a TDI array that makes use of both of these approaches is a 64×2048 -element array reported by Bradley and Ibrahim [40]. The TDI registers are designed for eight-phase ripple clocking. In this design, seven pixels, each associated with one charge packet, are created in each group of eight phases. The eight-phase ripple-clocked structure is constructed using two levels of polysilicon gates and self-aligned implanted barriers. Given the design rules that limit the dimension per phase to $12 \mu\text{m}$, this device has a Nyquist limiting resolution of $(8/7)(12 \mu\text{m}) = 13.7 \mu\text{m}$. The readout register uses the same structures (except that it has $13 \mu\text{m}/\text{phase}$) connected as a two-phase register. In a twofold interlace mode of clocking, charge packets are read out at a density of one every $13 \mu\text{m}$.

Bradley and Ibrahim [40] also report the development of a chip-bumping capability, which allows them to position several TDI chips end-to-end in an hermetic package with only a very small dead space between the active areas. In their design, the dead space is nominally $26 \mu\text{m}$, which is equivalent to two columns of photoelements. Successful mechanical fabrication and assembly are reported.

Additional design concepts for high-density TDI arrays are reported by Angle *et al.* [41]. These include the use of multiple readout registers in parallel with each other.

3. Performance

Thompson [42] reports the performance of a 9×20 -element TDI array designed for the NASA multispectral scanner. It utilizes tin oxide gates to achieve a quantum efficiency of approximately 65% from 450 to 850 nm. At a sample rate of 125 kHz, the gated charge-integrator-type preamplifier is linear over a dynamic range of 1000:1 to within 0.5%. A noise figure for the device was not reported.

Dyck [43] reports the performance of a 64×1024 -element TDI array designed for the long range electrooptical reconnaissance system (LOREORS). The device features high MTF and high signal-to-noise ratios near saturation. A four-phase TDI register design and a $20 \times 20\text{-}\mu\text{m}$ pixel size are used to give the device a nearly maximum MTF insofar as TDI clocking is concerned and to give it a saturation charge of 10^6 electrons/pixel, providing high signal-to-noise ratios near saturation.

Another large TDI array is a 128×1024 -element device designed for the low light level panoramic TDI charge-coupled (LOPATCH) device system [44]. This device features high quantum efficiency and a very low noise preamplifier. By control of the polysilicon thicknesses as well as silicon oxide and silicon nitride thicknesses so as to optimize broadband responsivity [45], a quantum efficiency of greater than 50% was achieved from 550 to 800 nm. The low noise preamplifier has a noise equivalent input signal of approximately 20 electrons/pixel rms.

4. Designs for High-Speed Page Scanning

Conventional CCD line-scan image sensors can, as we have seen, operate to approximately 20 MHz/output amplifier. For page scanning with a resolution of 200 lines/in. across an $8\frac{1}{2}$ -in. page, this leads to an integration period of approximately 100 μsec . This speed also appears to be approximately the practical limit set by illumination power that must be applied to the paper to give the desired signal and still not create a risk of overheating the paper. Thus, for higher speeds, TDI arrays with several parallel output amplifiers can be the solution. Angle *et al.* [41] illustrate several layouts for doing this with four amplifiers and describe developmental efforts to achieve some of them. Their specific goal is a combined output rate from the chip of 84 MHz.

III. PtSi-SILICON SCHOTTKY-BARRIER INFRARED IMAGE SENSORS

This review of CCD imagers has thus far dealt only with the intrinsic photoresponse in silicon. Schottky-barrier photodiodes are of interest, within the silicon semiconductor industry, for image sensors in that much more of the infrared spectrum becomes accessible while avoiding the materials problems of different semiconductor systems. The platinum silicide Schottky barrier on *p*-type silicon is attractive because its 0.27-eV barrier height provides response to 4.6- μm wavelength, which is well into the 3–5- μm atmospheric window. This window is particularly useful for passive infrared imaging applications.

Kosonocky *et al.* [46,47] report some quite exciting results from their development work on such imagers. Using a 25×50 -element array, as well as a linear array, and operating the arrays at temperatures of 50 to 90 K, they achieve noise equivalent temperatures (NET) as low as 0.4°C in a 26°C ambient and at a standard television frame rate. The quantum efficiency near the 4.6- μm cutoff, where the useful response occurs with 300 K radiation, is only of the order of 0.1%, but the 33 msec integration time and the low-noise CCD readout contribute heavily to the good results. In fact, test results lead these workers to predict a NET of 0.1°C with this type of imager.

A significant feature of this technology is the high uniformity of response that is possible in the case of illumination from the silicon side. This is important because passive imaging is inherently a low-contrast imaging problem. For example, the 0.4°C NET in a 300 K ambient corresponds to a theoretical noise equivalent contrast level of approximately 1%. The reported nonuniformity of response for this device was found to be as low as 0.5% rms. With this uniformity, useful imagery can be obtained without the need for pixel-by-pixel correction.

IV. OTHER INFRARED IMAGERS

For solid-state imaging applications involving longer wavelengths, a number of approaches are being investigated. Several of these approaches were reported at the 1978 International Conference on the Application of Charge Coupled Devices (San Diego). Chapman *et al.* [48] report HgCdTe CCD arrays for the 3–5- μm range that operate well at 77 K. A number of details of the fabrication are given; for example, the compositional nonuniformity of the recrystallization process was typically ± 0.3 mole%. As for

the possibility of HgCdTe CCD arrays for the 8–12- μm range, a number of materials problems are discussed, but the possibility is not ruled out. Barrowcliff *et al.* [49] report the successful operation of an infrared CCD imager fabricated in GaInSb. Thus a variety of different semiconductor systems are being investigated for infrared imagers.

V. PROSPECTS FOR THE FUTURE

A. High-Resolution Facsimile Scanners

Now that line-scan image sensors are available with a resolution of 2000 elements/device, a logical question regards the feasibility of manufacturing devices that will provide 300- or 400-line/in. resolution across an $8\frac{1}{2} \times 11$ -in. page, i.e., that will have up to 3400 or 4400 elements. Charge-transfer efficiencies required for such devices already have been achieved, namely, >0.99999 [50], but it would be undesirable to double the present chip length of ≈ 25 mm for mechanical reasons. Therefore, the unit cell probably needs to be scaled down by approximately a factor of two, from a typical value of $13 \mu\text{m}$ to the order of $8 \mu\text{m}$. The optical cross talk at an 8- μm element size, assuming no changes in vertical dimensions, would result in approximately a 2% decrease in MTF in the visible spectral range. This would be quite manageable. Thus, there do not appear to be any fundamental obstacles.

B. High-Speed Scanners for Electronic Mail

As discussed in detail by Angle *et al.* [41], this application requires a 1728-element or a 2200-element array with a total output sample rate of either 66 or 84 MHz. Because of the resulting short line-integration time, TDI imaging will apparently be required. The challenge is to partition the array so as to provide several parallel outputs so that each output can operate below approximately 20 MHz. Again, there does not appear to be any fundamental obstacle.

C. One-Chip Color TV Imagers

One-chip color TV imagers are, without question, the most interesting challenge considered here. The question is no longer whether such a chip can be made (see Section II.C.6). The question is rather if or when the cost of such a chip can be brought to the level needed to develop a large

(consumer) market. To try to predict cost from the number of masks and the yield per mask would merely give another prediction, which time would prove wrong. The business incentive is high, perhaps comparable with that of any recent new generation of integrated circuit memory. Therefore, a meaningful prediction might be sought by relating the color imager to some size of memory. The TV imager has some 200,000 elements. Current chip areas are approximately 100 mm². A competitive 256K DRAM probably will have one-third to one-half of this area. To decrease the area of the imager by even a factor of two would force a proportional reduction of the input signal level for a given scene illumination. This would require a comparable increase in quantum efficiency, a comparable decrease in the limiting noise (fixed-pattern or random), or a combination of the two. At this time it is not clear what fundamental limitation exists here other than the one of 100% quantum efficiency, and even this must be regarded carefully because it does not rule out the possibilities of gain in a phototransistor-type structure or in a photoconductive element. In August of 1981 a color TV camera that uses a single MOS sensor chip became available from Hitachi.

D. Color-Film Scanners for TV Display

If the requirement of a line-array color-film scanner for TV display is that it have 380 photoelements for each of three colors, then the total number of elements required is 1140. This is a relatively small number of photoelements. By the time that one-chip color TV imagers of the area-array type are in volume production, this scanner certainly could be a practicality, based simply on the fewer number of photoelements. The device might need to be of the TDI type instead of the simple line-scan type, but the number of TDI rows would probably not be large, and thus the argument based on relative complexity does not change.

E. "Smart Imagers" and "Smart Cameras"

A conventional image sensor scans an image, pixel by pixel, at megahertz rates, precisely "reading" the amplitude of the integrated photoelectronic signal from each pixel and thus generates image data at a very high rate. This high data rate can present several types of problems: (1) a heavy demand on recorder bandwidth, (2) a heavy demand on transmitter bandwidth, and (3) long computing times where the image data is to be analyzed or used for control purposes. A "smart imager" is considered to be an image sensor that alleviates any of these problems in the focal plane; a "smart camera" is one that does this in close proximity to the focal plane.

It is not the intent here to distinguish between these two or to analyze the general problem but rather to point out some recent efforts and developments in the direction of alleviating the above problems. The subject relates intimately to the fields of image analysis and image processing, on which much has been written in recent years. However, there appears to be little written that shows how some of the various image processing operations can be implemented at the focal plane.

One paper that reports a focal plane implementation is by Hall and Awtry [51]. It describes a 324×340 -element CCD image sensor having on the same chip a CCD delay line with nine independent taps placed so that at any instant a 3×3 submatrix is accessed. Hall and Awtry go on to show examples of imagery processed for (1) edge enhancement, (2) Laplacian edge detection, and (3) gradient image enhancement. Since their typical charge-transfer efficiency was 0.99997, they state that the 648-plus additional stages of CCD register for the image-processing function contributed a negligible signal degradation in most applications. Such a group of parallel outputs from a submatrix could also be used to facilitate delta-modulation algorithms for bandwidth compression.

A CCD signal-processing device that could appropriately be used in a camera for image processing is considered by Armstrong and McVey [52] for application in spacecraft navigation and camera pointing. The device is an analog correlator. A specific implementation of the correlator discussed is one described by Mavor *et al.* [53]. Their device employs single-FET four-quadrant multipliers. The correlator is used in the image-processing application to determine the position of an object subject to translation in three dimensions.

In the future, it would appear that such image-processing functions near the focal plane will greatly enhance the practicality of these kinds of systems.

F. Impact of New Fabrication Technologies

1. Wafer Stepping

For today's major line-scan imager, the 1728-element device for page scanning, commonly designed to $4\text{-}\mu\text{m}$ rules and having an active length of 22.4 mm, wafer stepping could bring about a reduction in size by perhaps a factor of two. This should provide a substantial cost reduction. The active length could then be 11.2 mm and the chip length approximately 13 mm. This length is still on the large side for typical wafer steppers, but stepping is still a definite possibility.

For television-type imagers, the wafer-stepping technology appears to offer a powerful advantage. The preferred sensor array size is 6.6×8.8

mm, which is a good size for wafer stepping. Furthermore, the design rules that become feasible with wafer stepping allow some very attractive possibilities in design, especially with respect to maximizing the photo-sensitive fraction of the area of each element.

2. Other Technologies

As shown in this chapter by many examples, there is a large assortment of technologies in use today, any one of which might become a major factor in new imager development and/or manufacture. One that has not yet been mentioned here but which has the potential for simplifying some imager structures is the edge-etch technology first reported by Hosack and Dyck [54]. "Edge etch" refers to a process for forming submicrometer cuts in a layer of material, the cuts being along the edges of a photolithographic pattern such that the cut width is independent of the resolution of the photolithographic process. Applied to a conductive layer of polysilicon in a frame-transfer imager, for example, the complete gate structure could be formed all in one process step. The advantages would include (1) simplicity of process, (2) optimum transmittance relative to structures with overlapping polysilicon layers, and (3) low interelectrode capacitance.

Without doubt, many possibilities still exist for improved solid-state imagers and, more specifically, for CCD imagers.

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